



Suicide Attempts in Juvenile Bipolar Versus Major Depressive Disorders: Systematic Review and Meta-Analysis

Franco De Crescenzo, MD, Giulia Serra, MD, Francesco Maisto, MD, Mai Uchida, MD, Hilary Woodworth, BA, Maria Pia Casini, MD, Ross J. Baldessarini, MD, Stefano Vicari, MD, PhD

Objective: Suicide attempts are prevalent in association with major mood disorders, and risk is greater with bipolar disorder (BD) than major depressive disorder (MDD) in adults. There may be similar relationships in juvenile mood disorders, but the evidence has not been compiled systematically and quantitatively.

Method: We searched for reports of studies comparing rates of suicide attempts in children or adolescents diagnosed with BD or MDD, and applied random-effects meta-analysis.

Results: In 6 reports from 1995 to 2017, with 2,303 participants diagnosed with mood disorder from the United States and South Korea, aged 3 to 18 years, rates of suicide attempts differed significantly by diagnosis: BD

(31.5%) > MDD (20.5%) > hypomania or mania-only (8.49%). Risk of suicide attempts differed (BD > MDD) highly significantly by meta-analysis (odds ratio [OR] = 1.71, CI = 1.33–2.20, $p < .0001$), and was very similar if a study with attempts and suicidal ideation was excluded (OR = 1.64, CI = 1.26–2.15, $p < .0001$).

Conclusion: Risk of suicide attempts in juvenile mood disorder patients ranked: BD > MDD >> hypomania or mania-only >> juvenile general population.

Key words: affective, bipolar, depression, meta-analysis, suicide attempt

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Suicide is the second leading cause of death worldwide among adolescents, and more than 80% of juveniles who have attempted suicide have had a major affective disorder.¹ In adults, suicide attempts are significantly more frequent among those with a diagnosis of bipolar disorder (BD) than of major depressive disorder (MDD).² However, among juveniles, the relative risk of suicidal behaviors with specific mood disorder types has not been addressed systematically.

Juvenile mood disorders are prevalent; they increase risk of suicide, substance abuse, and co-occurring medical illnesses, and they present high rates of hospitalization and potentially severe and disabling morbidity.^{3–5} Longitudinal studies have found that the prognosis of juvenile mood disorders are highly unfavorable, and that their diagnosis and treatment are often delayed for more than a decade, especially BD.^{5,6} Such delays probably reflect the difficulty of diagnosing BD at an early age, often starting with subsyndromal depressive or hypomanic symptoms, mood instability, mixed features, an ultrarapid cycling course, and occurrence of fully symptomatic hypomanic or manic episodes, with a clear manic-depressive episodic pattern only later. Also, in a large proportion of cases, the first major mood episode in BD is depressive, making the diagnosis of BD and differentiation

from MDD at an early age even more difficult and the use of potentially risky antidepressant treatment greater.^{5,6}

Abundant evidence in adult patients supports the conclusion that suicide attempts are more prevalent among persons diagnosed with BD compared to MDD. In a recent meta-analysis of rates of suicide attempts among adults with a mood disorder in 101 international studies, the pooled incidence was 31.1% (CI = 27.9–34.3), or 4.0% per year.⁷ Risk was similar among patients diagnosed with types I and II BD, and 2 times higher in both than in MDD.⁷ Among juveniles as well as adults, suicide attempts are strongly associated with major mood disorders, although relative associations with BD versus MDD remain uncertain.^{8–10}

Reported risk factors for suicide attempts in juvenile mood disorder patients include relatively young onset, female sex, severe depression, functional impairment, previous suicide attempts, nonsuicidal self-injurious behaviors, and a family history of suicidal acts.^{11,12} Among juveniles diagnosed with MDD, the presence of psychotic symptoms and insomnia,^{12,13} as well as “subthreshold” bipolar-like features, including hypomanic symptoms, cyclothymic or hyperthymic temperament, and prominent novelty-seeking behaviors^{14,15} also have been associated with suicide attempts. In addition, “mixed” manic-depressive features, including irritability, psychomotor agitation, or racing thoughts with dysphoria have strongly and consistently been associated with suicidal behaviors in adults, and there are indications of similar associations among juveniles with mood disorders.^{11,16–18} Given the large amount of evidence demonstrating a higher risk of suicide attempts in BD versus



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MDD in adults, and given the strongly suggestive evidence that subsyndromal hypomanic features may increase the risk of suicide attempts in juvenile MDD, we propose that, among juveniles, the risk of attempting suicide is preferentially associated with bipolar features, and that the relative risk of suicide attempts may be greater with BD versus MDD. Moreover, we propose that the clinical importance of earlier recognition of BD in juveniles and its differentiation from MDD includes the need to monitor particularly closely for emerging suicidal risk.

Given this background and the remaining uncertainty about the relative risk of suicide attempts among children and adolescents diagnosed with BD versus MDD, we carried out a systematic review of the research literature on this topic. We used meta-analysis to test quantitatively for differences in suicide attempts among juveniles with these diagnoses, as well as among patients found to have hypomania or mania without depression, in an effort to evaluate the contributions of bipolar tendencies to suicidal risk in young persons.

METHOD

Literature Search

We searched PubMed/MEDLINE, CINAHL, ISI Web of Science, and PsycINFO research literature databases to identify relevant, peer-reviewed articles on suicide attempts in children and adolescents with mood disorders. The search syntax was based on combinations of the terms: (bipolar disorder OR bipolar depression OR hypomania OR mania) AND (major depressive disorder OR depression) AND (juvenile OR child* OR adolesc* OR pediatric OR youth) AND (suicid* OR suicide attempt OR suicide plan OR suicide threat OR suicide gesture OR self-harm OR self-injur*). With no lower date limit, the search was continued to June 27, 2017. We also examined references cited in included reports as well as abstracts of recent psychiatric research conferences, and gray literature via Internet searches to find additional studies. The full search strategy is available in Supplement 1 and Table S1, available online. This systematic review accorded with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.¹⁹

Study Selection

We included reports of studies involving adolescents or children diagnosed with a major affective disorder (BD or MDD) by standardized criteria up to age 18 years, with at least one suicide attempt at any time. We considered a suicide attempt to be any self-injurious behavior with a non-0 level of intent to die.^{20,21} We excluded the following: review articles, editorials, letters, and comments; case reports; and diagnosis not based on standardized criteria.

Two experienced investigators (G.S., M.U.) independently reviewed titles and abstracts of the retrieved articles and applied the study inclusion and exclusion criteria. These investigators then independently reviewed the full-text articles to confirm eligibility of studies for inclusion. Disagreements were resolved by discussion to consensus.

Data Extraction

For each included study, 5 authors (F.D.C., F.M., G.S., M.U., R.J.B.) collected basic information (authors, reference citation, year, country) and study characteristics, including research design, diagnostic methods, diagnoses, participant counts, proportions by sex, age range and mean, rates of suicide attempts, definition of suicide attempt, and methods of evaluating attempts. The primary outcome measure was

the proportion of participants with at least one suicide attempt among those diagnosed with BD versus MDD in the same study. Secondly, we also compared other participants with hypomania or mania only ("mM") without clinically manifest depression. Two authors (F.D.C., F.M.) independently assessed the quality of methods and potential sources of bias for each study, based on the Newcastle–Ottawa Scale,²² scored as the percentage of items considered to be of high quality among the scale's 8 characteristics.

Data Analysis

We considered the number and proportion of individuals reported as having at least one suicide attempt, and compared rates of persons making suicide attempts between different mood disorders using random-effects meta-analysis. Reported analyses were carried out independently by 3 investigators (F.D., G.S., R.J.B.) to consensus. Meta-analytic findings are reported as pooled odds ratios (OR) with 95% CIs, based on STATA.13 statistical software (StataCorp, College Station, TX). Nonquantitative data are presented descriptively. We performed a sensitivity analysis by excluding the only study²³ that included suicidal ideation as well as suicide attempts. We assessed statistical heterogeneity in each pairwise comparison with the I^2 statistic derived from Q values provided by meta-analysis. We considered I^2 values as indicating very low (<10%), low (10%–25%), moderate (26%–50%), or high (75%–100%) heterogeneity.²⁴ We also performed meta-regression analysis to determine associations of reported study characteristics (including mean age, proportion of females, reporting year, participant count, and study quality rating) on the primary meta-analytic contrast of suicide attempt rates among juveniles diagnosed with BD versus MDD.

RESULTS

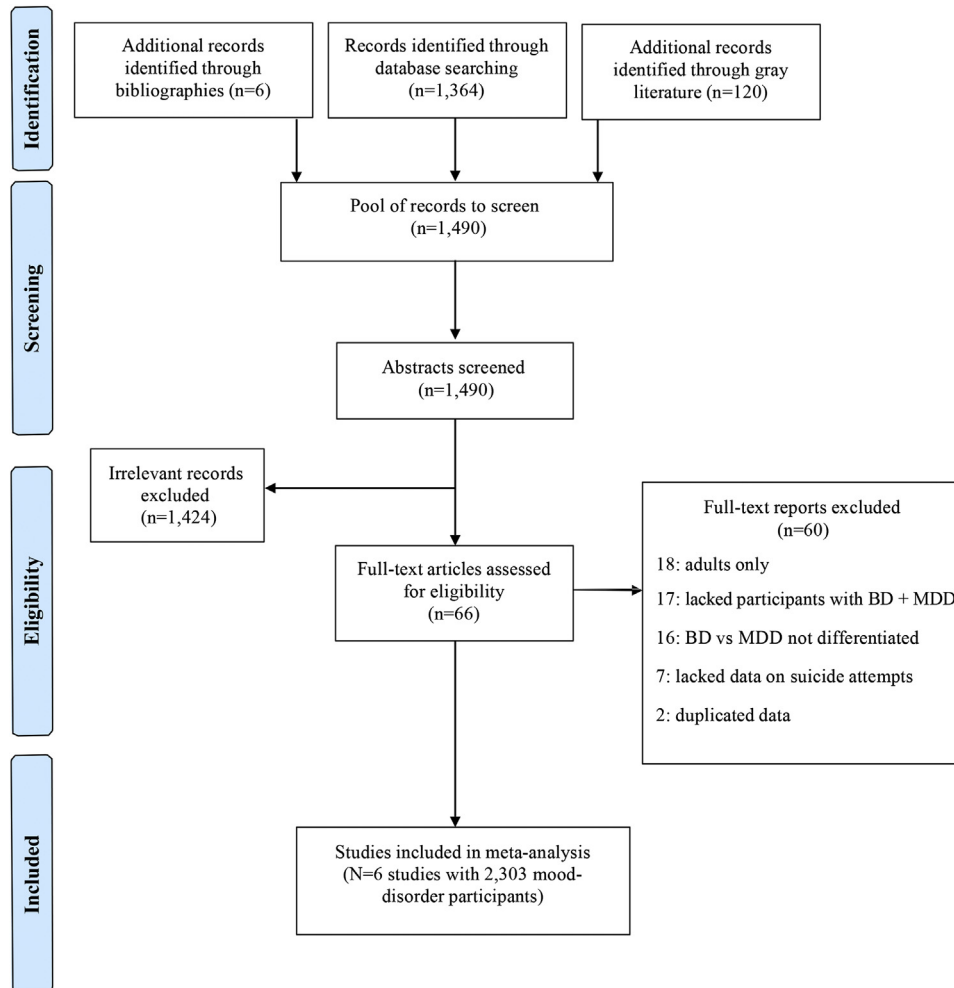
Studies Selected

The literature search generated 1,364 potential records, plus 120 from the gray literature and 6 from bibliographies of initially identified reports, leading to 1,490 records to be screened (Figure 1). Of these, 1,424 records not including information on required diagnostic categories or outcomes were excluded, leaving 66 full-text reports to be assessed for eligibility. Of these, 60 were excluded for the following reasons: included adults not separated from juveniles ($n = 18$); did not include both BD and MDD cases ($n = 17$); did not differentiate BD versus MDD cases ($n = 16$); lacked data on suicide attempts ($n = 7$); or duplicated data ($n = 2$). This process yielded 6 reports (with 2,303 participants diagnosed with a mood disorder) for analyses (Figure 1).^{16,23,25–28}

Study Characteristics

Salient characteristics of included studies are summarized in Table 1. Their diagnostic assessments were based on structured or semi-structured diagnostic interviews, in some cases backed with validated rating scales, following *DSM-III* to *DSM-5* criteria for BD and MDD. All included studies compared participants with BD and MDD. Two involved community samples of US high school students and included individuals with hypomanic symptoms and possible unipolar mania who were not included among those diagnosed with BD and were compared separately to those with BD or MDD in the same studies.^{25,27} Another study surveyed children diagnosed with putative BD versus unipolar MDD according to parental reports of symptoms and behavioral abnormalities.²⁶

FIGURE 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart of literature search to obtain 6 reports for analysis. Note: BD = bipolar disorder; MDD = major depressive disorder.



Four studies were cross-sectional^{23,26,28} or based on chart review,¹⁶ and 2 studies included prospective follow-up to 12.0²⁷ or 13.8 months.²⁵ Participant ages averaged 15.0 (CI = 12.8–17.2) years and ranged from 3 to 18 years (Table 1).

Four studies considered only patients reported to have made at least one suicide attempt,^{16,25,27,28} a fifth considered suicide attempts and suicidal ideation together,²³ and a sixth included suicidal and other self-injurious behaviors.²⁶ Quality ratings of the studies averaged 77.1% (CI = 61.8–92.4) of the maximum attainable score with the Newcastle–Ottawa Scale (Table 1).²²

Quantitative Analyses

Based on meta-analytically pooled values of reported rates of suicide attempters, participants diagnosed with BD were 1.54 times more likely to have made at least one suicide attempt than those with MDD (31.5%/20.5%) (Table 1). This difference was strongly supported by random-effects meta-analysis (odds ratio [OR] = 1.71, CI = 1.33–2.20; $z = 4.18$, $p < .0001$) (Table 2, Figure 2), with low between-study heterogeneity ($I^2 < 10\%$) (Table 2). This diagnostic difference in suicide

attempt rates remained highly significant when a study including suicidal ideation²³ was excluded (OR = 1.64, CI = 1.26–2.15; $z = 3.62$, $p < .0001$) (Table 2). We also estimated the number needed to harm (NNH) to differentiate the risk of suicide attempt in participants with BD versus MDD (NNH = 11.0, CI = 7.38–21.5), based on the reciprocal of the meta-analytically computed rate difference (1/RD) for suicide attempters between the diagnostic groups.

Two of the studies^{25,27} supported a tentative meta-analytic comparison of suicide attempts between individuals with BD or MDD versus a separate group of individuals diagnosed with only hypomania or mania (mM). Such participants were highly significantly less likely to attempt suicide than those with MDD (OR = 2.14, CI = 1.33–3.44; $z = 3.15$, $p = .002$), and even less than others diagnosed with BD but having episodes of depression (OR = 4.68, CI = 2.45–8.94; $z = 4.68$, $p < .0001$) (Table 2).

DISCUSSION

This appears to be the first systematic review and meta-analysis to compare risks of suicide attempts in children

TABLE 1 Characteristics of Studies Included in Meta-Analyses

Study, Year, Reference	Sample, Assessment	Quality Score ^a	Age (y) ^{b,c}	Diagnoses (n)	Females (%)	Diagnostic Criteria	Suicidal History and Definition of Attempt	Suicidal Behavior (n, %)
Lewinsohn <i>et al.</i> , 1995 ²⁵	Community; N = 1,276 prospective 13.8 mo	87.5	4–18 16.6 ± 1.2	18 BD 316 MDD 97 mM 845 HC	BD: 67 MDD: 70 (mM: 57) HC: 48	K-SADS and LIFE	“Have you ever tried to kill yourself or done anything that could have killed you?”	BD: 8/18 (44.4%) MDD: 70/316 (22.2%) mM: 14/97 (14.4%) HC: 10/845 (1.18%)
Wozniak <i>et al.</i> , 2004 ²³	Clinical; N = 152 cross-sectional	75.0	6–17 12.8 ± 3.5	43 BD+ADHD 109 MDD+ADHD	BD: 35 MDD: 51	K-SADS-E	Self- or parent-reported suicide ideation or attempt	BD: 29/43 (67.4%) MDD: 52/109 (47.7%)
Dilsaver <i>et al.</i> , 2005 ¹⁶	Chart review N = 229	75.0	14.7 ± 1.5	82 BD-Mx 147 MDD	BD: 44 MDD: 47	Chart review	Chart review Attempt: “a physically self-destructive act associated with the idea of terminating one’s life”	BD: 42/82 (51.2%) MDD: 68/147 (46.3%)
Luby and Belden, 2008 ²⁶	Community; N = 80 cross-sectional	62.5	3–6	26 BD 54 MDD	BD: 86 MDD: 69	PAPA-1.4	Parent-reported suicidal behavior	BD: 7/26 (26.9%) MDD: 8/54 (14.8%)
Merikangas <i>et al.</i> , 2012 ²⁷	Community; N = 1,213 prospective 12 mo	100	13–18	246 BD 805 MDD 162 mM	BD: 68 MDD: 68 (mM: 55)	CIDI-3.0	Self-reported suicide attempt	BD: 48/246 (19.5%) MDD: 102/805 (12.7%) mM: 8/162 (4.94%)
Shon <i>et al.</i> , 2014 ²⁸	Clinical; N = 198 review of 16 mo	62.5	6–18 15.1 ± 2.4	55 BD 143 MDD	BD: 54 MDD: 46	DSM-IV and COBY	Suicide attempt in medical record	BD: 14/55 (25.5%) MDD: 22/143 (15.4%)
Means ^b /Totals [95%CI]	6 Studies N = 3,148	77.1 [61.8–92.4]	15.0 [CI = 12.8–17.2]	470 BD 1,574 MDD (259 mM)	BD: 58.0 MDD: 63.7 (mM: 55.5)	---	---	BD: 148/470 (31.5%) MDD: 322/1574 (20.5%) (mM: 22/259 [8.49%])

Note: ADHD = attention-deficit/hyperactivity disorder; BD = bipolar disorder; BD-Mx = mainly mixed-states; CES-D = Center for Epidemiological Studies Depression Scale (self-reported); CIDI = Composite International Diagnostic Interview; COBY = Course and Outcome of Bipolar Youth study criteria for unspecified BD; D = depression; HC = healthy controls; K-SADS-E = Epidemiologic version of Children’s Schedule for Affective Disorders and Schizophrenia; LIFE = Longitudinal Interval Follow-up Evaluation; MDD = major depressive disorder; mM = manic or hypomanic without major depression; PAPA = Preschool Age Psychiatric Assessment; SHBQ = Self-Harm Behavioral Questionnaire.

^aNewcastle–Ottawa Scale quality assessment score is shown as percentage of 8 characteristics considered to be of high quality (sampling method; sample size; controls; defined exposure; age matching; sex matching; outcome assessment; and statistics).

^bAges of diagnostic subgroups did not differ from within-study means.

^cMeans are weighted by N. For pooled rates: BD vs. MDD ($\chi^2 = 23.0$); MDD vs. mM ($\chi^2 = 21.8$; both $p < .0001$).

TABLE 2 Summary of Random-Effects Meta-Analyses

Comparison	Studies	OR [95% CI]	NNH [95% CI]	I ²	z Score	p Value
BD vs. MDD (all reports)	6	1.71 [1.33–2.20]	11.0 [7.38–21.5]	<10%	4.18	<.0001
BD vs. MDD (attempts only)	5	1.64 [1.26–2.15]	12.4 [7.90–28.9]	<10%	3.62	<.0001
BD vs. mM	2	4.68 [2.45–8.94]	6.01 [4.41–9.47]	<10%	4.68	<.0001
MDD vs. mM	2	2.14 [1.33–3.44]	12.9 [8.56–26.5]	<10%	3.15	.002

Note: Studies and data included are shown in Table 1. Interstudy heterogeneity (I²) was low. Meta-regression analysis found no association of outcome with study size, proportion of females and males, participant age, reporting year, or study quality ratings (all $p \leq .80$; not shown). BD = bipolar disorder; MDD = major depressive disorder; mM = hypomania or mania without depression (not included with BD); NNH = number needed to harm with MDD vs. BD.

and adolescents with different types of mood disorders. Based on a systematic literature search, we identified and pooled data from 6 studies meeting study criteria, with both BD and MDD cases observed under the same conditions. Juveniles with BD had a highly significantly higher risk of suicide attempt than those with MDD (OR = 1.64–1.71) (Table 2), and both participants with BD (31.5%; OR = 4.68) and MDD (20.5%; OR = 2.14) had higher rates than among those with hypomania or mania without major depression (8.49%) (Tables 1 and 2). The ratio of suicide attempter rates among adults with BD versus MDD is reported to be approximately 2.4,^{2,29} or greater than was found in juveniles in the present study (1.64–1.71-fold) (Table 2). Nevertheless, juvenile major depressive episodes sometimes are later found to have mania or hypomania to support a diagnosis of BD, at rates ranging from 9.0% to 43%.^{6,30} This tendency to change from apparent MDD to BD in juveniles might lead to underestimating the ratio of suicide attempt rates among

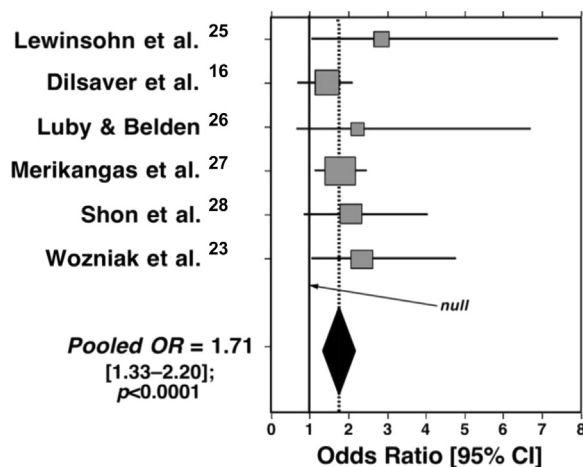
juveniles with BD versus MDD by inflating the rate among apparent MDD cases.

Rates of suicide attempts reported here are far higher than in general population samples of children and adolescents. We found only one report that included a rate of suicide attempts among healthy control juveniles, at 1.18% (CI = 0.57–2.17; 10/845) (Table 1).²⁵ Previously reported rates of suicide attempts between ages 9 and 18 years have ranged from about 3.3% to 7.0% (average ~5.2%) and to be more than twice higher among girls than among boys.^{9,31} Also important, the ratio of attempts to suicides (A/S, a proposed index of lethality of attempts) has averaged approximately 1,000 in girls but only 300 among boys of this age group in the general population, with similarly high A/S ratios for adults in the general population.^{8,9} Among adult patients diagnosed with a major affective disorder, the A/S ratio is far lower than in the general population, averaging approximately 10 in MDD and 8 in BD, indicating far greater lethality in mood disorder patients than in the general population.²⁹ In general, in both juveniles and adults, risk of suicidal behavior appears to rank as follows: general population << unipolar mania or hypomania << unipolar depression < bipolar disorder. Suicidal risks appear to be at least as great in highly depression-prone bipolar II as in bipolar I,^{2,29} and in cases of BD with prominent mixed manic-depressive states, at least in adults,² with less certainty about relative risks among juveniles with particular subtypes of BD.^{16,32}

The significance of the consistently higher risk of suicide attempts in juveniles and adults diagnosed with BD than with MDD remains to be clarified. The especially high suicidal risks associated with prominent manic-depressive mixed states, marked by agitation and increased energy and activity with prominent dysphoria, as well as the far higher risks among patients with BD with depressive as well as manic or hypomanic episodes as we found here, may offer some clues. For example, the combination of depressed or dysphoric mood with increased tendency to act, especially impulsively (as found in BD, including in manic-depressive mixed states), may provide particularly lethal conditions for suicide. The very low rate of suicide attempts found among young patients with mania or hypomania without obvious depression (Table 1) would seem to indicate the importance of a depressive or dysphoric component of pathological moods for risk of suicidal behavior.

The present preliminary findings in juveniles presenting mainly with mania or hypomania in 2 studies involving 259

FIGURE 2 Forest plot of results of random-effect meta-analysis of 6 reports comparing risk of suicide attempts in children or adolescents diagnosed with bipolar or major depressive disorder. Note: Heterogeneity was negligible ($I^2 = < 10\%$, $p = .66$), but random modeling was used. Pooled odds ratio (OR) = 1.71 (CI = 1.33–2.20, $z = 4.18$, $p < .0001$). Sensitivity analysis based on omitting the one study²⁹ that included participants with suicidal ideation as well as suicide attempts yielded a similar pooled OR = 1.64 (CI = 1.26–2.15; $z = 3.62$, $p < .0001$).



participants^{25,27} call for specific comment. Such cases were associated with much lower rates of suicide attempts (8.49%, CI = 5.40–12.6) than in BD (with depression as well as mania or hypomania; 31.5%, CI = 27.5–35.9) or MDD in the present study (20.5%, CI = 18.5–22.5) (Table 1), and only slightly higher rates than in the juvenile general population (8.49% versus 5.20%).^{8,25,26} Moreover, most juveniles presenting with early mania or hypomania without evidence of clinical depression have not developed fully expressed syndromal BD in later years.^{6,32–35} These findings underscore the complexity of recognizing and evaluating the range of affective disturbances in children and adolescents and their prognostic implications for later morbidity and suicidal risks.^{6,30}

The present findings have important clinical implications concerning early identification of BD, particularly in children and adolescents, as they encourage close monitoring of patients with juvenile mood disorders for potential suicidal risk. Of particular importance is uncertainty about the future course of illnesses presenting as depression in children and adolescents. Many cases of BD present with early depression and may be considered as MDD but later may be identified as BD by the emergence of hypomanic, manic, or mixed states, sometimes during treatment with an antidepressant or other mood-elevating medicine.³⁵ Major efforts are being made to improve early prediction and identification of BD in young people.⁶ Clinical indications of increased risk of future emergence of BD include behavioral excitation, labile mood, dysphoria, or anger and insomnia, as well as a family history of BD.⁶

The present primary findings have several limitations. Notably, we found only 6 reports (representing a total of 2,303 participants with either BD or MDD) that met study selection criteria, and even these sometimes involved broad

definitions of mood disorder diagnoses or of suicidal status or parental reports rather than direct clinical observation, perhaps with risk of overemphasis on depression and underrecognition of hypomania or mania (Table 1). Also, variability in the assessment of suicide attempts with the use of retrospective information from chart reviews may well yield inaccurate rates, owing in part to recall bias. Nevertheless, although inaccuracies in identifying suicide attempts and in the clinical assessment of mania and depression may well occur across studies, they would be expected to arise at similar rates among both MDD and BD cases within studies. In addition, as exposure times probably varied greatly and are not defined in the studies analyzed, it was not possible to estimate time-adjusted rates (% per year). Moreover, as most of the studies included were observational and cross-sectional, long-term diagnostic and behavioral outcomes in later, more mature, years are not known.

Diagnosis of BD was associated with a higher rate of suicide attempts than was diagnosis of apparent unipolar MDD, and both were higher than in cases with mania or hypomania without depression in 6 reports involving a total of 2,303 children and adolescents diagnosed with a mood disorder. Additional studies are required to test the findings presented, to better evaluate the long-term significance of isolated mania or hypomania without depression in juveniles, and to assess the impact of mixed manic-depressive states in juveniles. Practical, clinical conclusions pertaining to the evaluation and treatment of juvenile patients include the need for close consideration of early-onset BD because of its sometimes-difficult differentiation from MDD and especially high risk of suicidal behavior. ☞



Clinical Guidance

- Juvenile mood disorders are highly prevalent psychiatric diseases with potentially severe and disabling morbidity and increased risk of suicide. The incidence of suicide attempts among adults diagnosed with a mood disorder is 31%, and 2.4 times greater among those diagnosed with BD than with MDD.
- In 6 reports with 2,303 participants diagnosed with mood disorders, aged 3 to 18 years, rates of suicide attempts were similar to those reported in adults and differed significantly by diagnosis: BD (31.5%) > MDD (20.5%) > hypomania or mania only (8.49%).
- Risk of suicide attempts by meta-analysis was 1.7 times greater among juveniles diagnosed with BD versus MDD (OR = 1.71, CI = 1.33–2.20, $p < .0001$).
- The present findings have important clinical implications concerning the need for close monitoring of patients with juvenile mood disorders for potential suicide risk, as well as early identification of BD, with accurate differentiation of juvenile BD from MDD.

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Drs. De Crescenzo and Maisto are with Institute of Psychiatry and Psychology, Catholic University of the Sacred Heart, Rome. Dr. Serra is with Child Psychiatry Unit, Bambino Gesù Children's Hospital, IRCCS, Rome, Harvard Medical School, Boston, MA, and International Consortium for Mood and Psychotic Disorders Research, Mailman Research Center, McLean Hospital, Belmont, MA. Dr. Uchida is with Harvard Medical School and Pediatric Psychopharmacology Program, Massachusetts General Hospital, Boston. Ms. Woodworth is also with Pediatric Psychopharmacology Program, Massachusetts General Hospital. Drs. Casini and Vicari are with Child Psychiatry Unit, Bambino Gesù Children's Hospital, IRCCS, Rome. Dr. Baldessarini is with Harvard Medical School and International Consortium for Mood and Psychotic Disorders Research, Mailman Research Center, McLean Hospital.

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Correspondence to Giulia Serra, MD, Child Psychiatry Unit, Department of Neuroscience, Bambino Gesù Children's Hospital, IRCCS, Rome, Italy; e-mail: giuliaserra@gmail.com

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SUPPLEMENT 1

PROTOCOL

Review Question(s)

To review evidence of differences in suicide attempts in children and adolescents diagnosed with bipolar disorder (BD) versus major depressive disorder (MDD).

Searches

PubMed/MEDLINE, CINAHL, ISI Web of Science, and PsycINFO databases will be searched to identify relevant articles on suicide attempts in children and adolescents with mood disorders. The search syntax will be based on combinations of the terms: (bipolar disorder OR bipolar depression) AND (major depressive disorder OR depression) AND (juvenile OR child* OR adolesc*) AND (suicid* OR suicide attempt). Databases will be searched from the date of inception. References of studies included and of major reviews/meta-analyses will be screened as well. We will contact investigators and relevant study authors, seeking information about unpublished or incomplete studies.

Types of Study to Be Included

Observational studies (either prospective, retrospective, or cross-sectional) on children or adolescents with a diagnosis of BD or MDD and assessing suicide attempts will be included. We will exclude editorials, letters, comments, conference proceedings, review articles, and case reports.

Participants/Population

The participants consist of patients aged up to 18 years, of both sexes, with a primary diagnosis of BD, mania, hypomania, or major depression. Most recent studies are likely to have used *DSM-5*,¹ *DSM-IV*,² or *International Classification of Diseases–10th Revision (ICD-10)*³ criteria. Older studies may have used *ICD-9*,⁴ *DSM-III*⁵/*DSM-III-R*,⁶ or other diagnostic systems. *ICD-9* does not use operationalized criteria because it has only disease names and no diagnostic criteria, so studies using *ICD-9* will be excluded. Studies using other research diagnostic criteria will be included. A concurrent primary or secondary diagnosis of another psychiatric disorder will not be considered as an exclusion criterion. Trials in patients with mood disorder and with a concurrent serious medical illness will be excluded.

Outcome(s)

Our primary outcome measure is to compare the ratio of suicide attempts between participants diagnosed with BD versus MDD. Secondarily, we will compare patients with subthreshold BD and hypomania or only mania without depression to participants with BD or MDD.

Data Extraction (Selection and Coding)

Two researchers will independently review titles and abstracts of retrieved articles, applying inclusion and exclusion criteria. Moreover, 2 reviewers will independently extract quantitative and qualitative data from the studies retrieved. We will design and use a structured data abstraction form to ensure consistency of appraisal for each study. Information extracted will include study characteristics (such as lead author, publication year), participant characteristics (gender, age range, age of onset, setting, diagnosis), number of individuals reported as having at least one suicide attempt per group, and scale or mean to measure suicide attempts.

Risk of Bias (Quality) Assessment

We will assess risk of bias in the included studies using the Newcastle-Ottawa Scale as a reference guide.⁷ Where inadequate details are provided, study authors will be contacted for further information. The quality assessment will be done by 2 independent raters. If the raters disagree, the final rating will be made by consensus with the involvement (if necessary) of another member of the review group.

Strategy for Data Synthesis

Data will be entered independently by at least 2 researchers into a data abstraction sheet and then analyzed by using STATA-13 (StataCorp, College Station, TX). We will perform a pairwise meta-analysis using the random effects model.⁸ We will compare the number of children and adolescents with BD versus MDD with at least one suicide attempt. Secondarily, we will compare the number of children and adolescents with subthreshold BD and hypomania or only mania without depression to participants with BD or MDD. We will report results as pooled odds ratios (OR) with 95% CIs. We will also perform a sensitivity analysis by comparing studies presenting pooled results on suicide attempts and suicidal ideation with those reporting specifically on suicide attempts. Heterogeneity between studies will be investigated by the I^2 statistic⁹ (we will consider I^2 “low” if <25%, “moderate” from 26% to 75%, and “high” if >75%) and by visual inspection of the forest plots. Skewed data and nonquantitative data will be presented descriptively.

Analysis of Subgroups or Subsets

We will perform if possible a meta-regression to check for associations of suicide attempts with age or gender.

CHANGES TO THE PROTOCOL

After the final analyses, the search strategy was implemented with the inclusion of the search terms “hypomania,” “mania,” “self-injury” and “self-harm,” and conference proceedings were screened, according to specific requests from the peer review. Below we report the fully implemented search strategy and the final search output.

Search Strategy

Search strategy: Medline/Pubmed

- 1.exp "bipolar disorders" /
- 2.'bipolar depression'.mp
- 3.hypomania.mp
- 4.mania.mp
- 5.'major depressive disorder'.mp
- 6.depression.mp
- 7.1 OR 2 OR 3 OR 4
- 8.5 OR 6
- 9.juvenile
- 10.child*.mp
- 11.adolesc*.mp
- 12.pediatric.mp
- 13.youth.mp
- 14.9 OR 10 OR 11 OR 12 OR 13
- 15.suicid*.mp
- 16.'suicide attempt'.mp
- 17.'suicide plan'.mp
- 18.'suicide threat'.mp
- 19.'suicide gesture'.mp
- 20.'self-harm'.mp
- 21.'self-injur*'.mp
- 22.15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21
- 23.7 AND 8 AND 14 AND 22

Search strategy: CINAHL

- 1.(MH "bipolar disorders+")
- 2."bipolar depression"
- 3."hypomania"
- 4."mania"
- 5."major depressive disorder"
- 6."depression"
- 7.1 OR 2 OR 3 OR 4
- 8.5 OR 6
- 9."juvenile"
- 10."child*"
- 11."adolesc*"
- 12."pediatric"
- 13."youth"
- 14.9 OR 10 OR 11 OR 12 OR 13
- 15."suicide"
- 16."suicide attempt"
- 17."suicide plan"
- 18."suicide threat"
- 19."suicide gesture"
- 20.'self-harm'
- 21.'self-injur*'
- 22.15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21
- 23.7 AND 8 AND 14 AND 22

Search strategy: ISI Web of Science

- 1.TS='bipolar disorders'
- 2.TS='bipolar depression'
- 3.TS='hypomania'
- 4.TS='mania'
- 5.TS='major depressive disorder'
- 6.TS=depression

- 7.#1 OR #2 OR #3 OR #4
- 8.#5 OR #6
- 9.TS=juvenile
- 10.TS=pediatric
- 11.TS=child
- 12.TS=adolescent
- 13.#9 OR #10 OR #11 OR #12
- 14.TS=suicide
- 15.TS='suicide attempt'
- 16.TS='self-harm'
- 17.TS='self-injury'
- 18.#14 OR #15 OR #16 OR #17
- 19.#7 AND #8 AND #13 AND #18

Search strategy: PsycINFO

- 1."bipolar"
- 2."bipolar depression"
- 3."hypomania"
- 4."mania"
- 5."major depressive disorder"
- 6."depression"
- 7.1 OR 2 OR 3 OR 4
- 8.5 OR 6
- 9.juvenile
- 10."child"
- 11."adolescent"
- 12."pediatric"
- 13."youth"
- 14.9 OR 10 OR 11 OR 12 OR 13
- 15.suicid*
- 16."suicide attempt"
- 17."suicide plan"
- 18."suicide threat"
- 19."suicide gesture"
- 20.'self-harm'
- 21.'self-injury'
- 22.15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21
- 23.7 AND 8 AND 14 AND 22

We searched for unpublished studies and gray literature via Internet searches on the following websites using the terms "suicide," "self-harm," "child," "adolescent," "bipolar," "hypomania," "mania," "depression," and "major depressive disorder."

- The New York Academy of Medicine (<http://www.greylit.org>).
- <http://www.opengrey.eu>.

We also used the following sources:

- References of the articles included.
- Conference proceedings in the last 5 years likely to contain studies relevant to the review. These included an unstructured search in Google Scholar, in the American Psychiatric Association (APA) website (<https://www.psychiatry.org/psychiatrists/search-directories-databases/library-and-archive>), and in the European Psychiatric Association (EPA) website (e.g. <http://www.epa-congress.org/2018/programme-submission/2017-e-posters#.WVTa8hTFkdc>).

TABLE S1 Numbers of Citations Retrieved by Each Database

Databases, Trial Registers, and Other Sources	Citations
Databases	
PubMed	773
Web of Science	328
CINAHL	169
PsycINFO	94
Total (databases)	1,364
Gray Literature	
http://www.greylit.org	84
http://www.opengrey.eu	36
Other sources	6
Total citations	1,490

Note: The last update of the search was performed on June 27, 2017. The searches performed in Google Scholar and the conference proceedings were not included in the count.

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